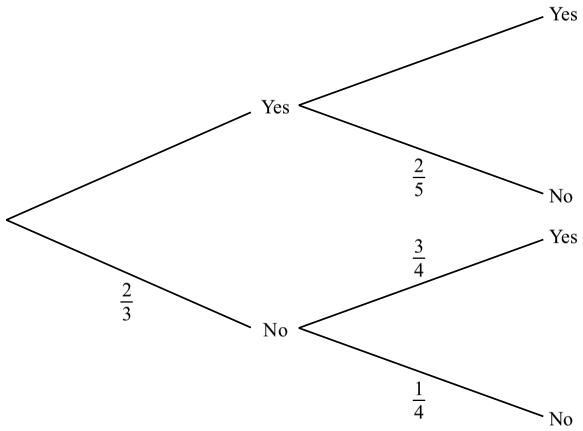
Question	Answer	Marks	Partial Marks
1	–2, –1, 0, 1	**2**	**B1** if one error/omission/extra
2	Correct triangle with arcs	**2**	**B1** correct no arcs
3	x^9y^4 final answer	**2**	**B1** for x^9y^k or x^ky^4 or correct answer seen then spoiled
4	–14, –5, 1	**2**	**B1** for 2 correct
5(a)	32.7 to 33.3	**1**	**FT** *their* printed diagram
5(b)	118 to 120	**1**	**FT** *their* printed diagram
5(c)	298 to 300	**2**	**B1** for 360 – (180 – *their* **(b)**) **FT** *their* **(b)** + 180
6(a)	5	**2**	**M1** for 10 × 12 × 'h' = 600
6(b)	172 or 172.4 to 172.5	**2**	**M1** for $1120 \times \frac{4}{3} \times \pi \times 0.45^3$
7(a)	20	**1**	
7(b)	 $\frac{2}{3}, \frac{2}{5}, \frac{3}{4}, \frac{1}{4}$ oe	**2**	**M1** for 2 or 3 correct
7(c)	$\frac{7}{10}$ oe	**3**	**M2** for $\frac{1}{3} \times \frac{3}{5}$ + *their* $\frac{2}{3}$ × *their* $\frac{3}{4}$ or **M1** for one product

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Question	Answer	Marks	Partial Marks
8(a)	$y = x$ ruled $x + y = 7$ ruled	3	**B1** for $y = x$ **B2** for $x + y = 7$ or **B1** for ruled line with gradient –1 or ruled line through (0, 7) or (7, 0) but not $x = 7$ or $y = 7$.
8(b)	Correct region below $y = x$, above $x + y = 7$ and above x-axis labelled R	1	
9(a)	$\frac{8}{3}$ oe	1	
9(b)	92	2	**M1** for correct calculation of an area
10	$(x + 3)(2 – 3y)$	2	**M1** for $2(x + 3) – 3y(x + 3)$ or $x(2y – 3) + 3(2y – 3)$
11(a)	6	2	**M1** for $10 + 13 – (20 – 3)$ oe or correctly completed Venn diagram
11(b)	7	1	**FT** *their* Venn diagram or 13 – *their* **(a)**
12	160.375	4	**M1** for mid-values for three of 125, 155, 162.5, 175 **M1** for Σfx where x is in correct interval including boundary **M1 dep** $\frac{\Sigma fx}{100}$ dep on second **M1**
13(a)	49.4 or 49.39 to 49.40	2	**M1** for $\tan = \frac{14}{12}$ oe
13(b)	22.4 or 22.36 to 22.37	4	**B1** for 62 + *their* **(a)** **M2** for $\sqrt{12^2 + 15^2 - 2 \times 12 \times 15 \cos(62 + \textit{their } \mathbf{(a)})}$ OR **B1** for 62 + *their* **(a)** **M1** for $12^2 + 15^2 – 2 \times 12 \times 15 \cos(62 + \textit{their } \mathbf{(a)})$ **A1** for 500.3 to 500.4
13(c)	13.2 or 13.24...	3	**M2** for $\sin 62 = \frac{\text{dist}}{15}$ or **M1** for recognition of shortest distance soi

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Question	Answer	Marks	Partial Marks
14(a)(i)	2400 2343.32	5	**B3** for 2400 or **B2** for answer 400 or **M1** for $\frac{2000 \times 2.5 \times 8}{100}$ oe **B2** for 2343.32 or **M1** for 2000×1.02^8 oe
14(a)(ii)	18.6 or 18.57 to 18.58...	2	**M1** for $\frac{\textit{their}\text{ total interest}}{4000}$ [× 100] or for $\frac{\textit{their}\text{ total amount}}{4000} \times 100$
14(a)(iii)	12	3	**M2** for trials below and above $n = 12$ or **M1** for at least 2 trials with $n > 8$
14(b)	2.8[0] or 2.799...	3	**M2** for $\sqrt[15]{\frac{7566}{5000}}$ or **M1** for $5000 \times [...]^{15} = 7566$
15(a)	35	2	**B1** for 49
15(b)	$\frac{y^2}{u^2}$ oe final answer	2	**M1** for $y^2 = u^2x$
16(a)	13.4 or 13.41 to 13.42	3	**M2** for $\sqrt{(-5-7)^2 + (8-2)^2}$ oe or **M1** for $(-5-7)^2 + (8-2)^2$ oe
16(b)	[y =] $2x + 5$ final answer	4	**M1** for [gradient of AB =] $\frac{8-2}{-5-7}$ oe **M1dep** for gradient of perpendicular, $p = -1 \div \textit{their} -\frac{1}{2}$ oe **M1dep** on previous **M1** for substituting (−1, 3) into $y = \textit{their}\ px + c$ oe where $\textit{their}\ p \neq 0$
17	0.427 or 0.4273 to 0.4274	2	**M1** for $\frac{1}{2} \times 13 \times 18 \sin x = 50$
18	$-\frac{1}{2}$ oe	3	**B2** for $12y = 6y - 3$ or better or **M1** for $4 \times 3y = 3(2y - 1)$

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Question	Answer	Marks	Partial Marks
19	391 or 391.1 to 391.2	4	**M3** for $3 \times 6 \times 20 + 2 \times \frac{1}{2} \times 6 \times 6 \times \sin 60$ oe OR **M1** for [3 ×] 6×20 **M1** for [2 ×] $\frac{1}{2} \times 6 \times 6 \times \sin 60$ oe
20	[k =] 9 [u =] 3	2	**B1** each
21	$\frac{5p}{2(p+4)}$ or $\frac{5p}{2p+8}$ final answer	3	**B1** for $5p(p-4)$ **B1** for $2(p+4)(p-4)$
22(a)(i)	$-\frac{1}{3}\mathbf{p} + \frac{1}{3}\mathbf{t}$ or $\frac{1}{3}(-\mathbf{p} + \mathbf{t})$ final answer	2	**M1** for $\overrightarrow{PT} = -\mathbf{p} + \mathbf{t}$ or $\overrightarrow{TP} = -\mathbf{t} + \mathbf{p}$
22(a)(ii)	$\frac{2}{3}\mathbf{p} - \frac{1}{3}\mathbf{t}$ or $\frac{1}{3}(2\mathbf{p} - \mathbf{t})$ final answer	2	**M1** for a correct route
22(b)	$\overrightarrow{OM} = \frac{2}{3}\mathbf{t} - \frac{2}{3}\mathbf{t} + \frac{4}{3}\mathbf{p}$ oe	M1	
	$\overrightarrow{OM} = \frac{4}{3}\mathbf{p}$ and $\overrightarrow{OM}$ is a multiple of $\overrightarrow{OP}$ oe	A1	
23	3.6	3	**M1** for $\frac{\textit{their}\text{ distance}}{\textit{their}\text{ time}}$ **B1** for 7.95 or 7950 or 132.5

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Question	Answer	Marks	Partial Marks
24	(–2.79, –4.58) (1.79, 4.58)	**6**	**B5** for (–2.791... , –4.583 to –4.582) **and** (1.791..., 4.582 to 4.583) or **B4** for –2.79 or –2.791... **and** 1.79 or 1.791... OR **M1** for $2x + 1 = x^2 + 3x - 4$ or better **M2** for $\frac{-1 \pm \sqrt{1^2 - 4(1)(-5)}}{2(1)}$ **FT** *their* quadratic not $x^2 + 3x - 4$ or **M1** for $\frac{-1 \pm \sqrt{p}}{2(1)}$ or $\sqrt{1^2 - 4(1)(-5)}$ **FT** *their* quadratic not $x^2 + 3x - 4$

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